

Smart AI Pharmacy Management System

Complete Operational Blueprint, Feature Architecture & Client Proposal

Document: System Proposal & Client Pitch Guide

Target: Retail Pharmacy Digital Transformation

Architecture: Standard PC + CPU AI + Cloud Sync

1. AI Vision & Biometric Automation

SMART EDGE AI

Entrance Footfall & Rush Counter

- **Real-time Customer Counter:** Dukan ke entrance par laga camera rozana aane walay tamam customers ko accurately count karega.
- **Peak Rush Hours Analysis:** Din ke kis hisse me sab se zyada rush hota hai (e.g. 7 PM - 10 PM) taake counter par staff pura ho.
- **Conversion Rate Tracking:** Agar 500 log aaye aur 320 bills banay, to 180 log kyun wapas gaye? (Dawai shortage ya slow service?).

Fingerprint Attendance & Payroll

- **Biometric Check-In/Out:** Staff ki attendance ungli ke nishan se lage gi (Buddy-punching aur fraud mukammal khatam).
- **Automatic Shifts & Overtime:** Late arrival, half-day, overtime aur shift timings ka automatic hisab.
- **Salary Report:** Mahinay ke aakhir me staff working hours aur salary sheet tayar.

2. Stock Intelligence & Supplier Automation

LOSS PREVENTION

Near-Expiry Early Warning (Critical)

- **30 / 60 / 90 Days Alerts:** Medicines expire hone se pehle alerts milenge taake unko discount par becha jaye ya company ko return kiya jaye.
- **Zero Expiry Waste:** Har maheenay hazaron rupe ki expired dawaiyon ka nuqsan 0% par le aayen.

1-Click WhatsApp Purchase Order

- **Auto Low-Stock Alert:** Jab kisi dawai ka stock safe limit (e.g. 5 dabba) se kam ho, system notification show karega.
- **Instant Order:** 1 click dabane par distributor ke WhatsApp par pre-typed purchase order message send ho jaye ga.

Dead Stock Identification

Pichlay 3 se 6 maheenay me jo medicines bilkul nahi biki, unki list taake dukan ka locked sarmaya (capital) aazad karwaya ja sake.

Existing POS Live Synchronization

Current desktop software ke database (SQL Server/Access/MySQL) se live stock aur billing data sync hota rahe ga.

3. Sales Boost & Customer Retention

REVENUE GROWTH

Smart Generic / Substitute Finder

Agar customer aisi dawai mangay jo stock me khatam hai (maslan *Panadol*), to system screen par fauran usi formula ke doosray brands dikha dega (maslan *Disprol*, *Calpol*). Grahak khali hath wapas nahi muray ga.

Chronic Patient Refill Reminders

Sugar, BP aur Heart ke mareezon ko 30 din ki dawai khatam hone se 3 din pehle automatic reminder: "*Aapki monthly medicines khatam hone wali hain, kya hum deliver kar dein?*" (Permanent customer retention).

WhatsApp Digital Invoicing

Paper receipt ke sath customer ke WhatsApp par direct bill bhejna. Paper bachat aur verified customer mobile directory tayar hona.

Controlled / Narcotics Drug Register

Neend aur dimaghi adwiyaat (Xanax, Lexotan etc.) kis doctor ke prescription par kis ko di gayi, uska regulatory compliance record.

4. Accounts, Udhaar & Remote Owner Control

REAL-TIME OVERSIGHT

Owner Mobile Live Dashboard

Pharmacy owner dukan par mojud na bhi ho, tab bhi mobile se dekh sake:

- Aaj ki total sale, gross margin aur profit
- Cash counter (Till) me is waqt kitna cash hona chahiye
- Dukan me kitne customer aaye aur staff attendance

Khata Book (Customer & Supplier)

- Distributor Ledger:** Kis company ka kitna udhaar baqi hai aur agla check kab dena hai.
- Customer Udhaar:** Mohallay ke regular gahakon ka hisab-kitab aur automatic WhatsApp payment reminders.

5. Technical Architecture & Hardware Requirements

NORMAL PC COMPATIBLE

Component	Recommended Specs	How it works on Normal PC
Host PC (System)	Core i5 (6th-8th Gen+), 8GB-16GB RAM, SSD	Baghair kisi mehngi Graphics Card (GPU) ke standard CPU par smoothly run karega.
Entrance Camera	Standard USB HD Webcam ya RTSP IP Camera	YOLOv8-Nano / YOLO11-Nano lightweight model via OpenVINO. 15-25 FPS real-time.
Fingerprint Scanner	DigitalPersona 4500 ya ZKTeco ZK4500	Hardware SDK / SourceAFIS minutiae matching. 99.9% accurate, instant (<0.1s), 0% CPU load.
Existing Desktop App	Current POS software already in use	Background sync utility database ko safely read kar k web dashboard ko feed kare gi.
Printer & Scanner	Thermal 80mm printer & 1D/2D Barcode scanner	Existing hardware as-is connect ho jaye ga.

6. Meeting Checklist: Questions to Ask Pharmacy Owner

DISCOVERY

Meeting me maloom karne wali ahem baten:

- Existing Software ka Naam aur DB:** Software ka naam kya hai? Peeche database konsa hai (SQL Server, MS Access, MySQL, SQLite)?
- PC Specifications:** Counter par paray PC ka processor, RAM, aur SSD/HDD kya hai?
- Distributor List:** Kitne distributors hain aur kya unke numbers/medicines ki list Excel me mojud hai?
- Staff Details:** Total kitne log kam karte hain aur unki shift timings kya hain?
- Entrance Point:** Shop ka entrance kaisa hai? Camera kahan fix kiya ja sakta hai (door ke theek ooper ya 45-degree angle)?
- Branding:** Pharmacy ka Logo aur theme colors (taake dashboard unhi colors me customized ban sake).

7. Client Pitch: 3 Golden Points to Win the Deal

CLOSING STRATEGY

1. Expiry Waste Ka Mukammal Khatma

"Har mahine hazaron rupe ki dawaiyan expire ho kar kachray me phainkna band. System 2 maheenay pehle alert dekar unko time pe clear karwaye ga."

2. Lost Sales Ka Hisab & Generic Finder

"Agar 100 customer aaye to 20 wapas kyun gaye? Footfall camera aur generic substitute feature se aapki sale foran 15-20% barh sakti hai."

3. Total Peace of Mind & Anti-Theft Control

"Aap dukan par na bhi hon, to ghar ya safar me baith kar mobile par live cash drawer, customer footfall, aur fingerprint staff attendance dekh sakein ge."